METRIC

MIL-DTL-83538/11A
8 July 2005
SUPERSEDING
MIL-DTL-83538/11
24 December 1998

DETAIL SPECIFICATION SHEET

CONNECTORS AND ACCESSORIES, ELECTRICAL, CIRCULAR, UMBILICAL, ADAPTER, BUFFER PLUG, NON-REMOVABLE PIN CONTACTS, FOR MIL-STD-1760 APPLICATIONS (METRIC)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-83538.

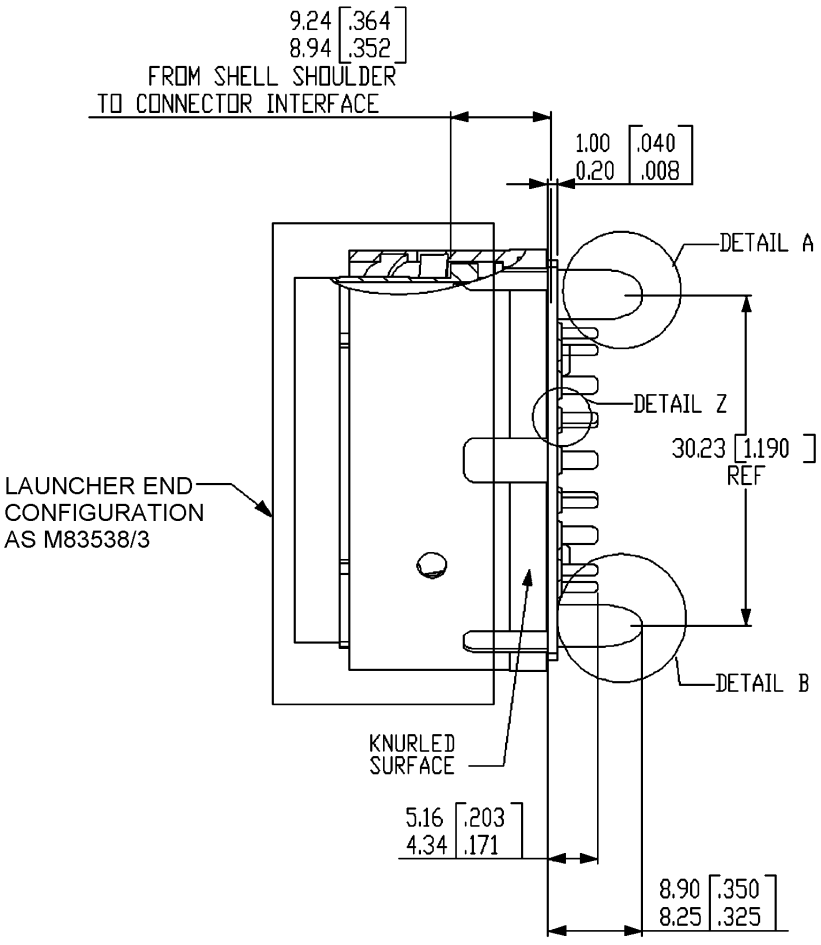


FIGURE 1. Buffer adapter plug, type 1.

AMSC N/A FSC 5935

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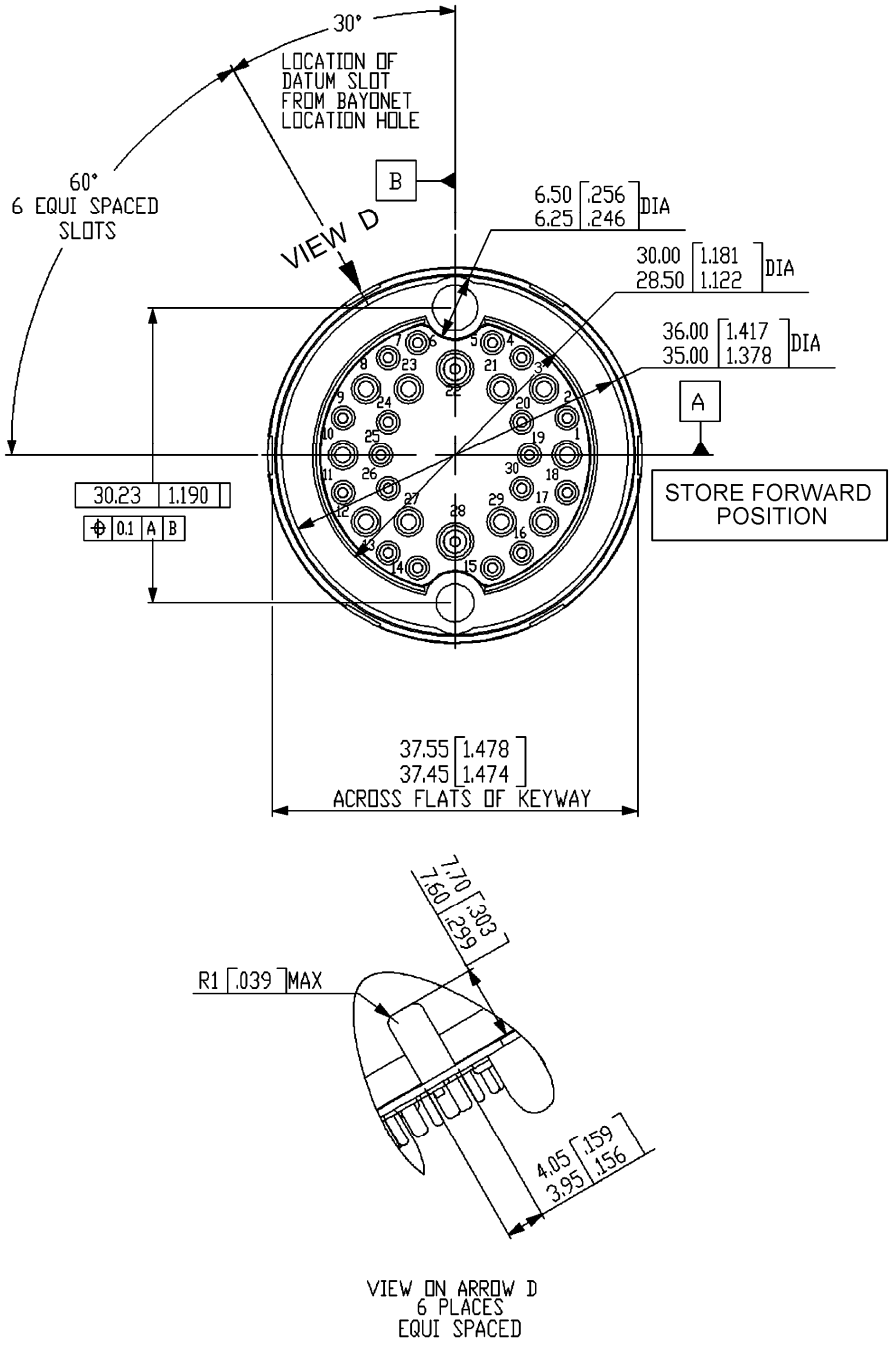
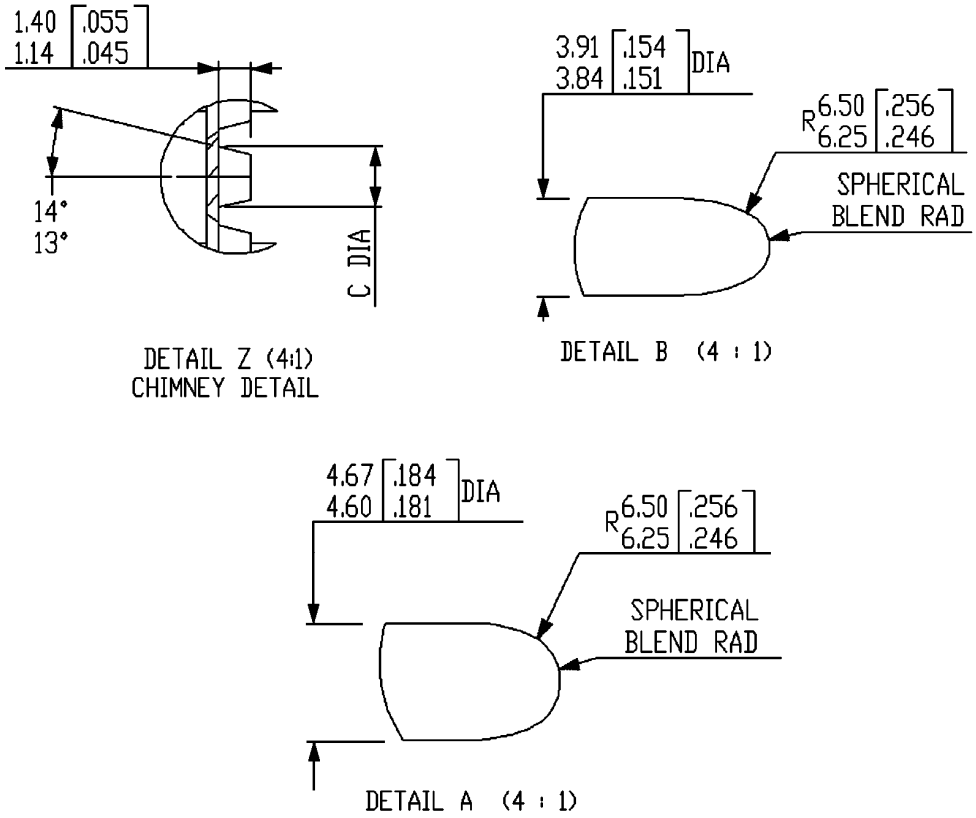


FIGURE 1. Buffer adapter plug, type 1 - Continued.

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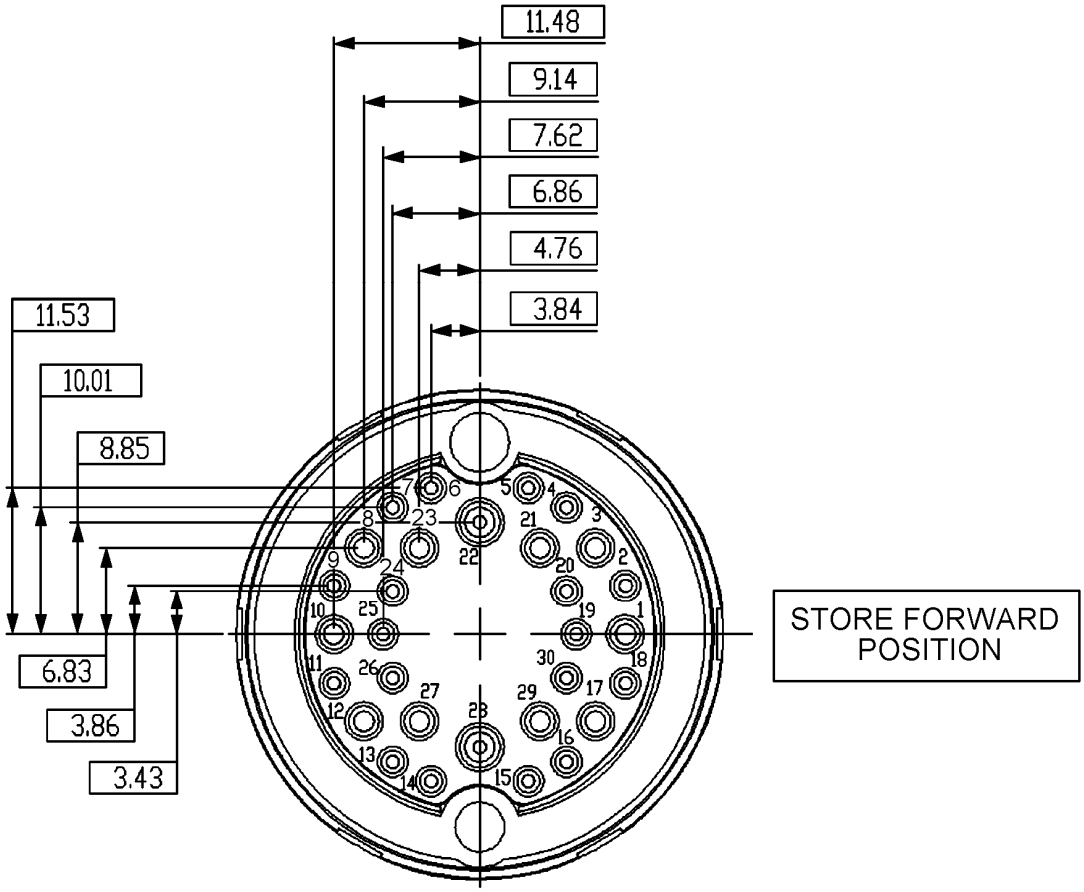


Contacts	C dia ± 0.1
20	2.41 (.095)
16	3.00 (.118)
12	3.81 (.150)

NOTES:
1. Dimensions are in millimeters.
2. Inch-pound equivalents are given for information only.

FIGURE 1. Buffer adapter plug, type 1 - Continued.

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mm	Inches	mm	Inches	mm	Inches
3.43	0.135	6.83	0.269	9.14	0.360
3.84	0.151	6.86	0.270	10.01	0.394
3.86	0.152	7.62	0.300	11.48	0.452
4.76	0.187	8.85	0.348	11.53	0.454

NOTES:
1. Dimensions are in millimeters.
2. Inch-pound equivalents are given for information only.
3. Hole positions are symmetrical about central axes.

FIGURE 2. Contact arrangement, type 1 - store interface.

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TABLE I. Pin connection details (Cross Reference Matrix Connection path). 1/

Aircraft Side (MIL-STD-1760)				Store Side (AIM120)	
Nomenclature	Type	Size	Contact	Contact	Size
270V DC Return	---	16	N	1	16
MUX BUS B-High	Twinax	8	H (center)	2	20
28V DC Power 2	---	16	F	3	16
LB (non-inverting)	Twinax	8	A (center)	4	20
LB (inverting)			A (intermed.)	5	20
Release Consent	---	20	1	6	20
LB (Shield)	Twinax	8	A (outer)	7	20
115V AC ∅ A	---	16	P	8	16
Address Bit A0	---	20	L	9	20
115V AC ∅ B	---	16	M	10	16
Address Bit A1	---	20	X	11	20
115V AC ∅ C	---	16	J	12	16
Address Parity	---	20	G	13	20
Address Return	---	20	6	14	20
Interlock Return	---	20	S	15	20
Interlock	---	20	B	16	20
28V DC Power 2 Return	---	16	E	17	16
MUX BUS A-High	Twinax	8	K (center)	18	20
MUX A & B Shield	Twinax	8	K & H (outer)	19	20
MUS BUS B-Low	Twinax	8	H (intermed.)	20	20
28V DC Power 1	---	16	C	21	16
				22	12 2/
Structure Ground	---	16	T	23	16
Address Bit A2	---	20	7	24	20
Address Bit A3	---	20	4	25	20
Address Bit A4	---	20	V	26	20
115V AC Neutral	---	16	Z	27	16
				28	12 2/
28V DC Power 1 Return	---	16	D	29	16
MUX BUS A-Low	Twinax	8	K (intermed.)	30	20
270V DC Power	---	16	R 2/		
Fiber Optics Channel 1	---	16	Y 2/		
Fiber Optics Channel 2	---	16	U 2/		
HB2	Coax	12	W 2/		
HB4	Coax	12	2 2/		
HB3	Coax	12	3 2/		
HB1	Coax	12	5 2/		

1/ Every effort has been made to connect contacts on the AIM120 side of the interface to their equivalent contacts on the MIL-STD-1760 side of the interface. It should be noted that, in order to satisfy certain store requirements, Pin 1 (AIM120 side) connects with Pin N (MIL-STD-1760 side).

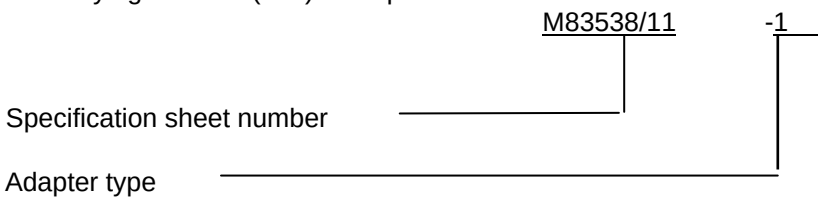
2/ These contact cavities shall not be used and contacts shall not be Installed in these cavities in the buffer. Their counter part cavities in the receptacle shall be properly sealed.

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REQUIREMENTS:

Dimensions and configurations: See figures 1 and 2 and table I.
This connector mates with MIL-DTL-83538/4 and AIM-120 store (or missile) receptacle.
For insert arrangement: See figure 2 and MIL-STD-1560 (25-20).

Part or Identifying Number (PIN) example:



Qualification: Connectors shall meet the qualification requirements of MIL-DTL-83538.

Referenced documents. In addition to MIL-DTL-83538, this document references the following:

MIL-DTL-83538/4
MIL-STD-1560
MIL-STD-1760

CONCLUDING MATERIAL

Custodians:
Army – CR
Navy – AS
Air Force – 11
DLA – CC

Preparing activity:
DLA – CC

Review activities:
Army – AR, MI
Navy – EC, MC, OS
Air Force – 19, 99

(Project 5935–4827–000)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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